

Cube Numbers

A cube number is a number multiplied by itself three times. Learn the cubes table, the digit patterns, the prime-factorisation test, and the "smallest number to multiply/divide for a perfect cube" problems.

What is a cube number?

The idea: Multiplying a number by itself **three times** gives a **cube number** (or **perfect cube**). We write it with a small raised 3.

$$5 \times 5 \times 5 = 5^3 = 125 \quad (\text{a cube of side 5 is made of 125 unit cubes})$$

The cubes you should know by heart:

n	1	2	3	4	5	6	7	8	9	10
n ³	1	8	27	64	125	216	343	512	729	1000

Also useful: $11^3 = 1331$, $12^3 = 1728$, $13^3 = 2197$, $15^3 = 3375$.

The patterns that get tested

- Last-digit rule.** Unlike squares, a cube can end in **any** digit 0–9. The units digit of n^3 depends only on the units digit of n : $1 \rightarrow 1$, $2 \rightarrow 8$, $3 \rightarrow 7$, $4 \rightarrow 4$, $5 \rightarrow 5$, $6 \rightarrow 6$, $7 \rightarrow 3$, $8 \rightarrow 2$, $9 \rightarrow 9$, $0 \rightarrow 0$.
- Trailing zeros come in 3s.** A perfect cube can only end in a number of zeros that is a **multiple of 3** (3, 6, ...). So a cube can **never** end in exactly two zeros.
- Sum of consecutive odd numbers.** $1 = 1^3$; $3 + 5 = 8 = 2^3$; $7 + 9 + 11 = 27 = 3^3$; $13 + 15 + 17 + 19 = 64 = 4^3$. The n -th group of odd numbers adds to n^3 .
- Perfect-cube test (prime factorisation).** Group the prime factors in **threes**. If every prime forms complete triples with none left over, the number is a perfect cube.

Smallest number to multiply / divide: Whatever primes are short of a triple, multiply by them to complete the triples (or divide to remove the extras) — a classic 3-mark question.

Famous fact — 1729 (the Hardy–Ramanujan number): it is the smallest number expressible as the sum of two cubes in two ways: $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

Prerequisite refresher: "three times" means $n \times n \times n$, not $n \times 3$. So $4^3 = 64$, not 12.

Where marks slip away

Trap 1 — Cube $\neq \times 3$. $4^3 = 4 \times 4 \times 4 = 64$, not 12. This is the single most common cube error.

Trap 2 — Using the square rule for cubes. Cubes can end in 2, 3, 7, 8 — the "forbidden digits" rule is for squares only.

Trap 3 — Grouping in pairs. For cubes you group primes in **threes**, not pairs. Pairs are for square roots.

Trap 4 — Two trailing zeros. A cube cannot end in exactly two zeros — zeros must come in multiples of three.

Slip 5 — Notation. Write the index as a small raised 3: 5^3 . Don't write "53".

Model answers — write them like this

Example A. Is 9 a perfect cube?

Step 1: List nearby cubes: $2^3 = 8$ and $3^3 = 27$.

Step 2: 9 lies between 8 and 27, so it is no whole number cubed.

$\therefore$ 9 is NOT a perfect cube

Example B. Find the units digit of the cube of 23.

Step 1: Only the units digit of 23 matters, which is 3.

Step 2: $3^3 = 27$, which ends in 7.

$\therefore$ Units digit of $23^3 = 7$

Example C. Is 3375 a perfect cube? Use prime factorisation.

Step 1: $3375 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$.

Step 2: Group in threes: $(3 \times 3 \times 3) \times (5 \times 5 \times 5)$.

Step 3: Both primes form complete triples, so 3375 is a perfect cube.

$\therefore$ 3375 is a perfect cube ($= 15^3$)

Example D. By what smallest number should 3600 be multiplied to make a perfect cube?

Step 1: $3600 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 2^4 \times 3^2 \times 5^2$.

Step 2: For a cube, every exponent must be a multiple of 3. Currently 2^4 (need 2 more 2s), 3^2 (need 1 more 3), 5^2 (need 1 more 5).

Step 3: Multiply by $2^2 \times 3 \times 5 = 4 \times 3 \times 5 = 60$.

Check: $3600 \times 60 = 216000 = 60^3$.

$\therefore$ Multiply by 60

Example E. By what smallest number should 128 be divided to make a perfect cube?

Step 1: $128 = 2^7$.

Step 2: The nearest lower multiple of 3 for the exponent is 6, so remove one 2.

Step 3: $128 \div 2 = 64 = 2^6 = 4^3$.

$\therefore$ Divide by 2 (quotient $64 = 4^3$)

Next: [try the worksheet](#), then [check the answer key](#).